

114 年度 社團法人中華產業機械設備協會 推廣費曼學習記錄

主講者 曾俊雄

日期 114 年 12 月 09 日

1.學習項目：鍋爐給水節煤器之設計要求。

2.學習動機：代檢員對設計法規其語意不明之處進一步了解及探討，避免因不熟稔法規而解釋錯誤，造成代檢機構與事業單位因見解不同，造成雙方有所爭執。

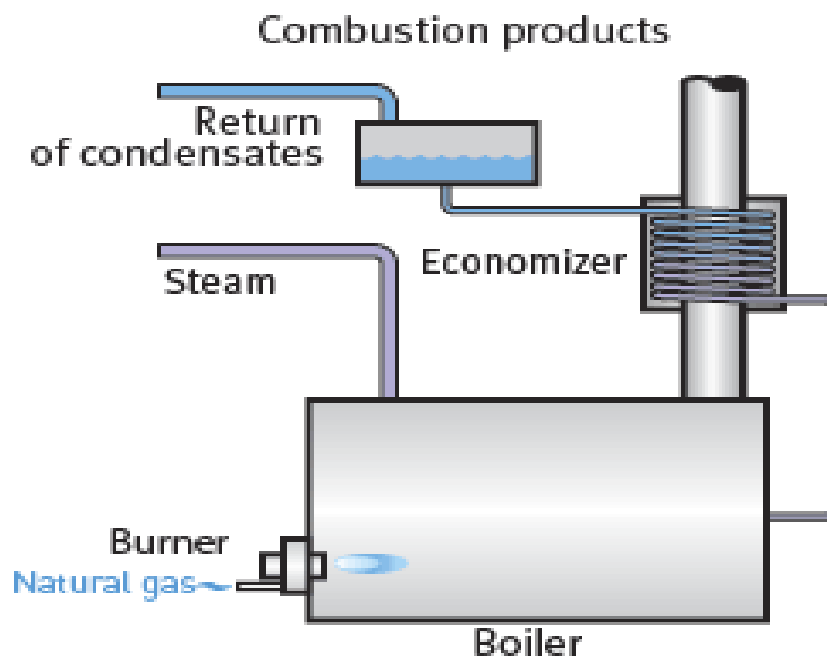
3.績效與效益：

- 1.藉由教育訓練讓代檢員熟知應如何查詢日本(JIS)及美國(ASME)鍋爐檢查標準比對節煤器之設計要求。
- 2.為增進代檢員理論上之本質學能：設計原理講述給水泵所產生之靜水頭壓是指靜止液體在重力作用下產生的壓力。就要考量泵浦的給水壓力及相對應的溫度，節煤器強度是否大於所能承受的來作計算。
- 3.對於節煤器、發生之腐蝕態樣進行圖像教育，增進檢查技術與技能。

學習相片剪影



ASME PFE REQUIREMENTS FOR FEEDWATER ECONOMIZERS (鍋爐給水節煤器之設計要求)



中華產業機械設備協會

代檢課長 曾俊雄

日期：114年12月9日

JIS B 8201 (2013)

6.8.12 円筒形管寄

6.8.13 長方形管寄

6.8.12 円筒形管寄せの最小厚さ 円筒形の管寄せの最小厚さは6.2.2による。ただし、附属設備としてのエコノマイザ用の場合は、 P は給水に差し支えない圧力とする。

6.8.13 長方形管寄せ 断面が長方形の管寄せは、次による。

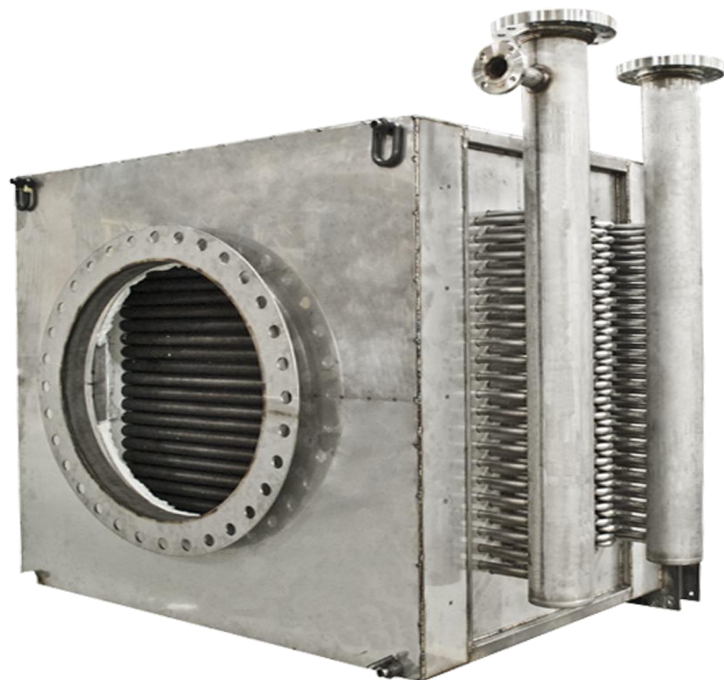
a) 四辺形断面のすみにおける内面の曲がりの半径は、管寄せの厚さの1/3より大きくし、**8 mm以上**としなければならない。

b) 直管部の最小厚さは、次の式によって求めた値のうちいずれか大きいほうをとる。ただし、内面の曲がりの半径が $m/5$ 以上であるときは、求めた値から10%を減じることができる。

P ：最高使用圧力 (MPa) 。ただし、附属設備としてのエコノマイザ用の場合は給水に差し支えない圧力。

節煤器集管器形狀

圓形集管器



非圓形集管器

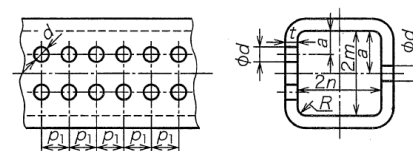


圖 29 長方形集管器

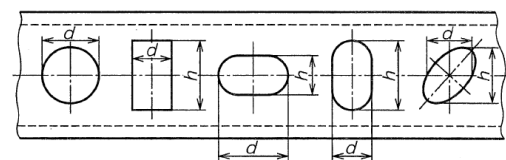


圖 30 集管器孔洞之尺度



2025/11/18 10:19

對節煤器的 P ：最高使用壓力 (MPa) 要求作比較

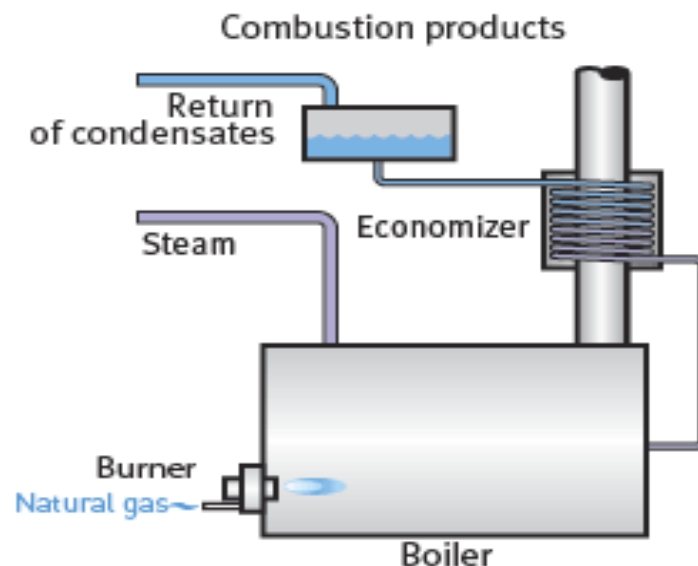
CNS 2139 (108年版) 6.8.13

P ：最高使用壓力 (MPa)
但，若係作為附屬設備之節煤器用時， P 為「**未妨礙給水**」之壓力。

重點：「未妨礙給水」改為
「不影響給水」(93年版
12.12~13)之壓力，舊版**原文**
可能較好理解。
如果將節煤器作為附屬設施，
 P 則壓力不得影響供水。

JIS B 8201 (2013) 6.8.13

P ：最高使用圧力 (MPa)
ただし，附屬設備としての
エコノマイザ用の場合は給
水に差し支えない圧力。



PFE REQUIREMENTS FOR FEEDWATER ECONOMIZERS

PFE-1 GENERAL

A feed water economizer is a **heat exchanger** in which **feedwater** to be supplied to a boiler is **heated by flue gases exiting** the boilers.

節煤器是一種熱交換器，用於利用鍋爐排出的煙氣加熱來供給鍋爐的給水。

When an economizer is provided with a **Section I boiler and resides** within the limits of boiler **external piping**, it **shall be** constructed in **accordance** with **Section I** rules.

當節煤器與SEC. 1鍋爐搭配使用，且位於鍋爐外部管道範圍內時，應依照SEC. 1的規定進行建造。

When the economizer is **located outside** the limits of **boiler external piping**, the economizer **shall be** constructed in **accordance** with the rules of **either Section I or Section VIII, Division 1**.

當節煤器位於鍋爐外部管道範圍之外時，應依照SEC.1或SEC.8 Div.1的規定進行建造。

REQUIREMENTS FOR FEEDWATER ECONOMIZERS

PFE-2 DESIGN (設計考量)

PFE-2.1 (壓力) The maximum allowable working pressure (MAWP) of the primary (feedwater) side of the economizer *shall be not less than (不小於) the design pressure requirements of ASME B31.1, para. 122.1.3(a)(1).*

When selecting the MAWP of the economizer, consideration (考量) *should be* given to the dead head pressure (靜水頭壓) of the feedwater pump as this pressure *may exceed* the value selected above.

註:靜水頭壓是指靜止液體在重力作用下產生的壓力。它會垂直且均勻地向所有方向施加，且壓力與深度成正比，深度越深，壓力越大。

PFE-2.2 (溫度) The design temperature of the tubes *shall be the greater* of either the saturated (飽和) steam temperature corresponding (對應) to the MAWP of the economizer *or* the maximum mean metal temperature (平均金屬溫度) under operating conditions.

Feedwater Piping(給水管路)

(a) The **value of P** to be used in *the formul* as in para.104 **shall be** as follows:

- (1) For piping from the boiler to and including the required stop valve and the check valve, the minimum **value of P** **except** as permitted **in (4) shall exceed the maximum allowable working pressure of the boiler by either 25% or 225 psi (1550 kPa), whichever is the lesser.**

P 的最小值應**超過**鍋爐最大允許工作壓力的 25% 或 225 psi (1550 kPa) , 取較小者。

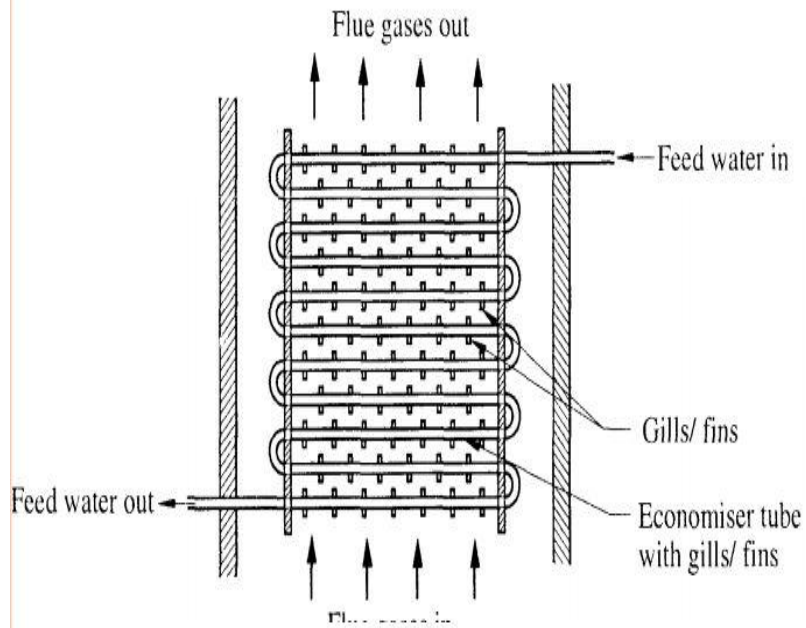
For an installation with an **integral economizer without valves between the boiler and economizer**, this paragraph shall **apply only** to the piping from the economizer inlet header to and including the **required stop valve and the check valve.**

對於有整體式節煤器且鍋爐和節煤器之間沒有閥門的裝置，本節僅適用於從節煤器入口集管到所需停止閥和止回閥（包括這些閥門）的管道。

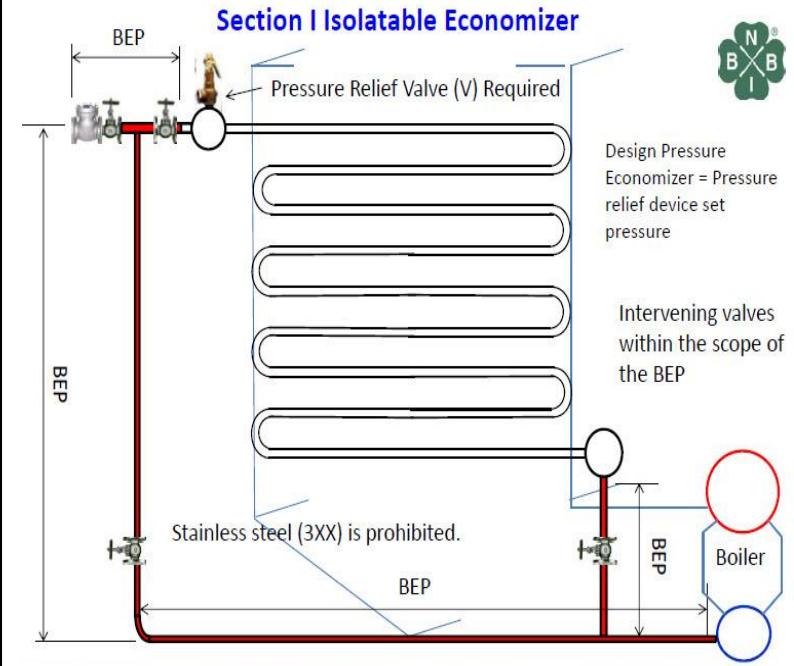
ASME Code Sec. I(隔離閥)

No intervening valve

Integral economizer



With intervening valve



[Section I Preamble, PG-5.5, PG-67.2.1.6 & PG-69.1.6]

10

Feedwater Piping

- (2) For piping between the required check (止回) valve and the globe or regulating (調節) valve, when required by para. 122.1.7(b), and including any bypass piping up to the shut off valves in the bypass, the value of P **shall be not less than the pressure required to feed the boiler.**
- (3) The value of P in the formula **shall not** be taken at **less than 100 psig [700 kPa (gage)]** for any condition of service or material, and **shall never be less than the pressure required to feed the boiler.**
給水管 P 則如下之規定，未滿0.7 MPa時，取為0.7 MPa。
(CNS 2139 6.8.7)

Feedwater Piping

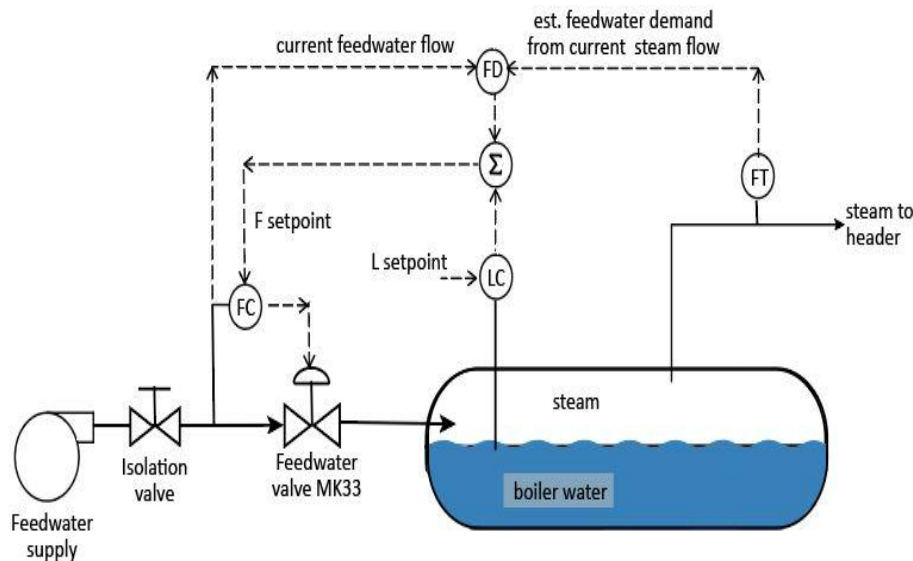
- (4) In a **forced-flow steam generator** with no fixed steam and water line, the value of P for feedwater piping from the boiler to and including the required stop valve may be in accordance with the requirements of para. 122.1.1(b).
- (b) The **S value** used, except as permitted in (a)(4), **shall not exceed that permitted for the temperature of saturated steam** at the maximum allowable working pressure of the boiler.
- (c) The size of the feed piping between the boiler and the first required valve [para. 122.1.7(b)] or the branch feed connection [para. 122.1.7(b)(4)] shall, as a minimum, be the same as the boiler connection.

PFE-4 REQUIREMENTS FOR FEEDWATER ECONOMIZERS

MATERIALS

When the feed water-**regulating valve** is located **between** the **boiler** and **economizer**, the economizer **may be** constructed **using austenitic stainless steel**, but materials **shall be limited** to those **listed** in **PG-12.3**.

當鍋爐與節煤器之間有**給水調節閥**時，節煤器可採用**沃斯田鐵系**不銹鋼製造，但材料應限於**PG-12.3**所列材料。



ASME Code Sec. I

PG-12.3

The material shall be one of the grades from the following list:

Chamber material of the remote **water level indicator**(遠程水位指示器) or water level-sensing devices(水位傳感), **except** for water columns(水柱), may include **austenitic stainless steels** and **nickel-based alloys**.

The material **shall be** in the **solution-annealed** (固熔)heat treatment condition. If **filler metals** are used in welding of the austenitic stainless steels, they shall be limited to **low carbon** content.

除水柱外，遠程水位指示器或水位傳感裝置的胴體材料，可包括沃斯田鐵系不銹鋼和鎳基合金。材料應處於**固溶退火狀態**。如果在其中使用填充金屬**沃斯田鐵系不銹鋼**的焊接，**應限制為低碳含量**。

Grade UNS Number

304L **S30403**

316L **S31603**

800 N08800

... N08020

825 N08825

C-276 N10276

... N06022

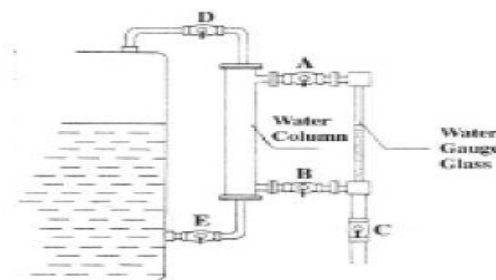
690 N06690

59 N06059

625 N06625

600 N06600

BLOWING OF WATER LEVEL GAUGE



A Typical Water Level Gauge

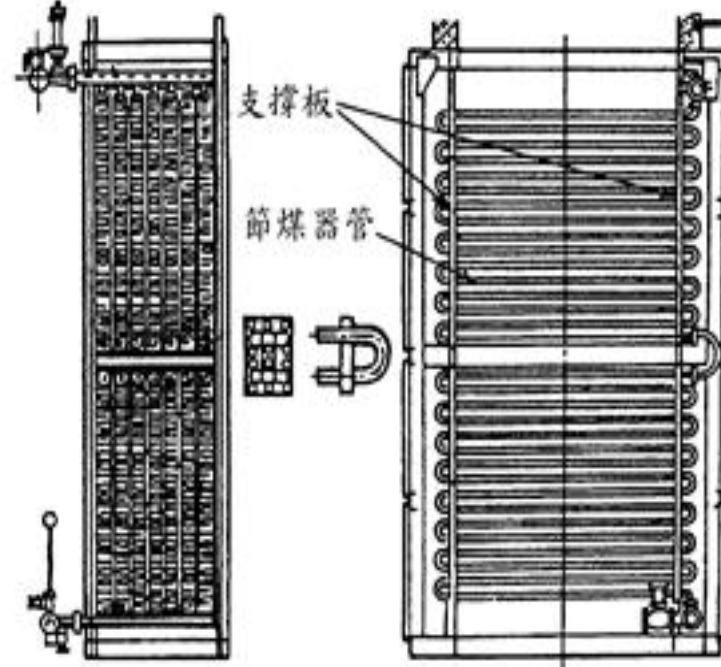
節煤器腐蝕原因

鍋爐檢查指引

節煤器接頭之漏水，會吸收煙道氣中之硫化物，而造成嚴重腐蝕。

依安檢則第86條之附屬裝置檢查規定，檢查節煤器（如右圖）有無腐蝕，有腐蝕時，則依照外部檢查水管之腐蝕處理方式。

其判定基準為：節煤器腐蝕，其殘厚已不能承受最高使用壓力者，判定不合格。



Description of Damage

Sulfuric acid promotes **general** and **localized corrosion** of carbon steel and other alloys. **Carbon steel HAZs** may experience **accelerated** .

硫酸會促進碳鋼和其他合金的**均勻腐蝕和局部腐蝕**。碳鋼的熱影響區可能會加速腐蝕。

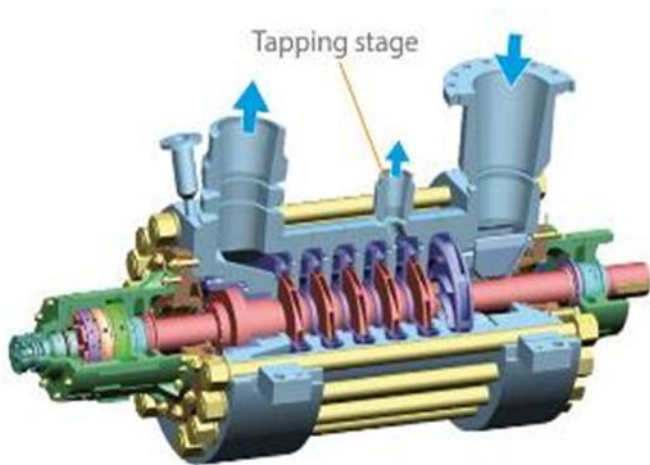


Figure 3-62-2—Accelerated sulfuric acid **corrosion under a baffle**.

節煤器結論：

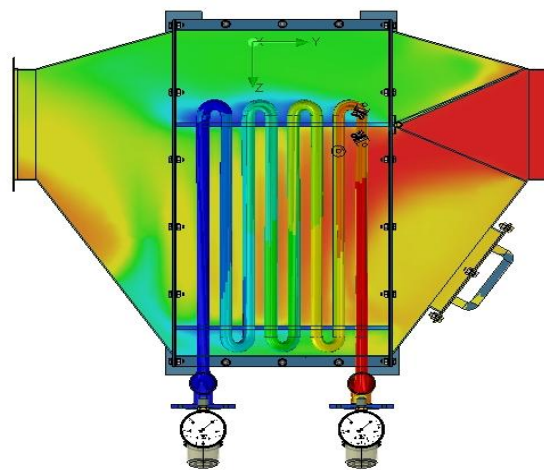
最高使用壓力(MAWP)

即給水壓力(所稱給水壓力：由鍋爐給水泵提供、而泵浦的轉速和設計也決定了給水壓力的大小)。



設計階段(Design)

就要考量泵浦的給水壓力及相對應的溫度，節煤器強度是否大於所能承受的來作計算。



課後評量：<https://forms.gle/tBWkDeDSwyFTKh4m6>

參考文獻資料	QR Code
1. ASME Code Sec.I 2. ASME Code B31.1 3. CNS 2139 4. JIS B 8201 5. API RP 571 6. 鍋爐檢查指引	